

Paperweight batch run analysis

Report directory: `/code/wg21-papergate/paperweight-out-v8`

Paperstore DB: `/code/wg21-paperflow/data/paperstore.db`

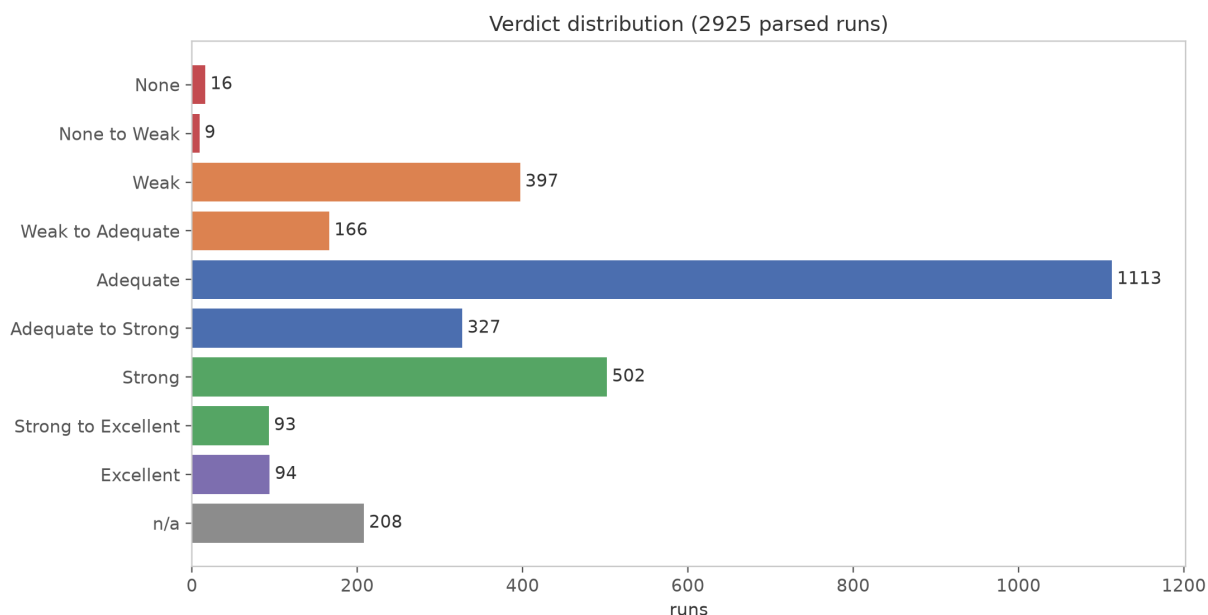
Batch overview

- **2925** report files across **585** papers (5 runs x 585 papers)
- expected: **2925** report files across **585** papers (5 runs x 585 papers, from paperstore DB)
- missing: **0** across **0** papers
- n/a verdicts: **208** (7.1% of parsed) — a triage turn that reads only the paper's front matter (title, abstract and anything before the first heading) classified it as not a proposal, so the criteria were not applied
- score (x/14) over 2717 scored runs: min 0, max 14, mean 6.05, median 6, stdev 2.81

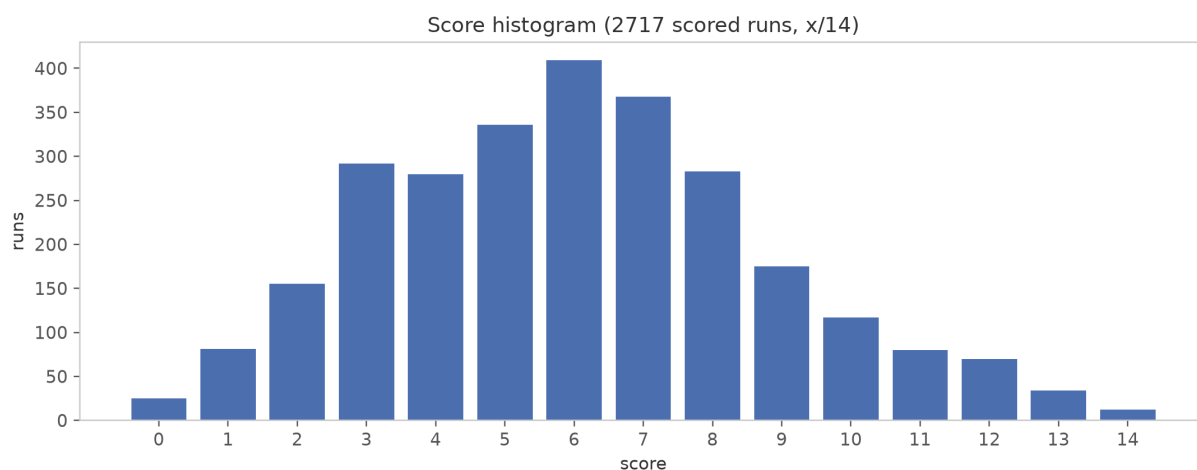
What the score is. Each run grades the paper on 7 criteria: why it matters, who is affected, prior art and alternatives, why the standard, coordination and interoperability, why a library will not do, and implementation experience. The paper is split into units, one per H2 section plus the front matter (sections over 30,000 characters are split into parts), and every criterion is graded on every unit by 3 independent samples: 0 (not addressed), 1 (asserted, with nothing supporting it) or 2 (supported with specifics). A non-zero grade only stands if it is backed by a verbatim quote of 3–40 words found in the paper, and bookkeeping sections (revision history, acknowledgements, references, poll records, wording) are graded 0 whatever they mention. A unit's grade is the mean of its 3 samples. A criterion's grade is the mean of its two best units, except implementation experience, which takes its single best unit, since one checkable pointer is enough. The score is the sum of the 7 criterion grades, a fraction from 0 to 14, and the report shows it rounded to the nearest integer. The verdict label is a fixed lookup on that integer: None = 0, Weak <= 3, Adequate <= 7, Strong <= 11, Excellent <= 14. **Span labels.** The run also computes the total each of the 3 samples would have produced alone. When those totals fall in different bands from the verdict's, the label becomes a span from the lowest band reached to the highest, e.g. "Adequate to Strong (7/14)". The number is still the verdict's own score, so it sits in one of the span's bands, not necessarily the first.

Verdicts and scores

Verdict distribution: how many runs received each verdict label. Plain bands and "X to Y" spans are separate bars, ordered top to bottom by the bands they cover, so each span sits between the bands it joins. A span is colored like its lower band. n/a is last.

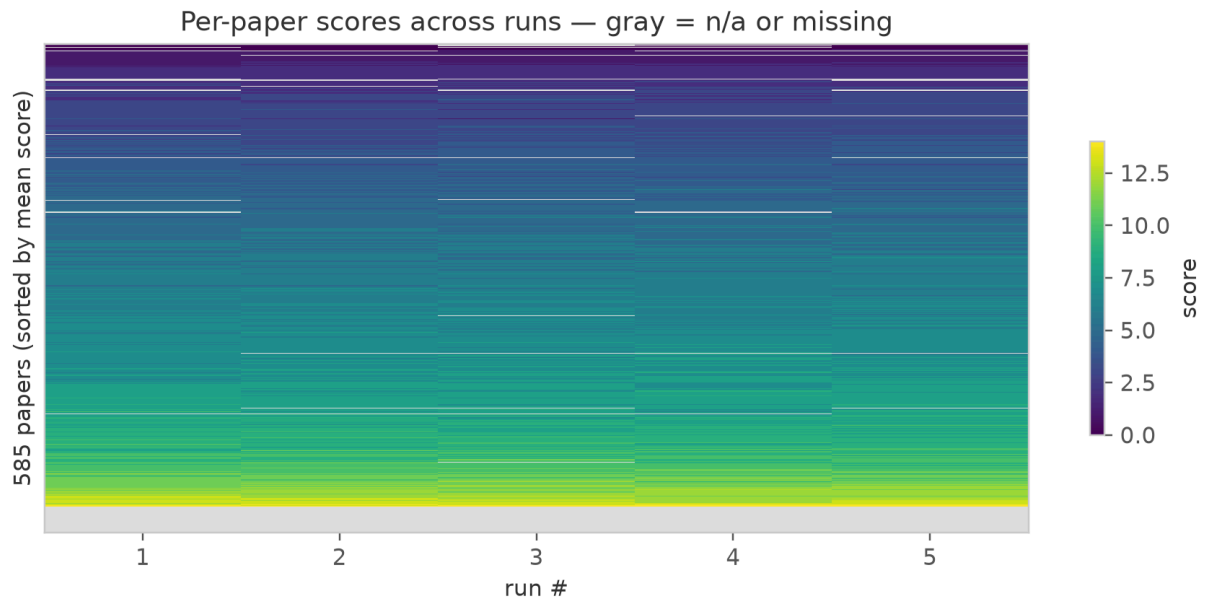


Score histogram: how many scored runs received each score. This is the rounded integer shown on the verdict line; n/a runs have no score and are not counted.



Scores across runs

One row per paper and one column per run, colored by the run's score. Papers are sorted by their mean score, lowest at the top. A row that changes color across runs is a paper whose score moves between runs; gray cells are n/a or missing runs.

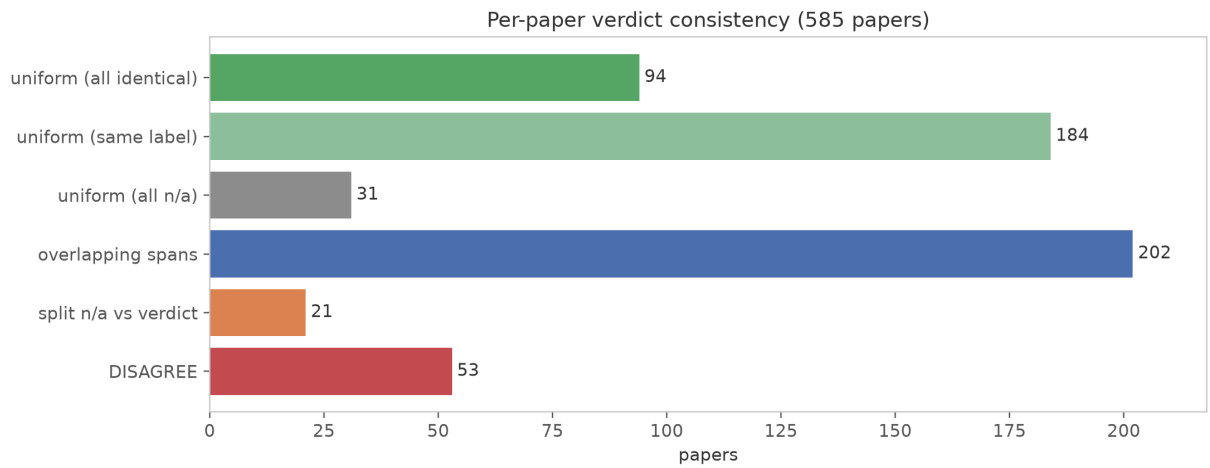


Consistency between runs

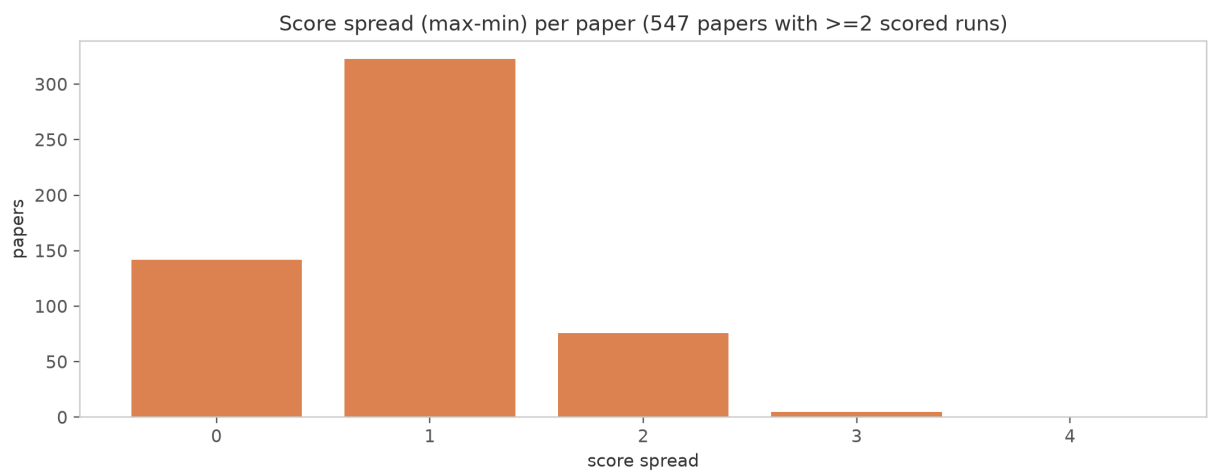
Every paper is run several times, and each paper is placed in exactly one category by comparing the verdict labels of its runs:

- **uniform (all identical):** every run has the same label and the same score.
- **uniform (same label):** every run has the same label, but the scores differ.
- **uniform (all n/a):** every run classified the document as not a proposal.
- **overlapping spans:** the labels differ, but there is at least one band that every run's label covers. For example, "Adequate", "Adequate to Strong" and "Weak to Adequate" all cover Adequate. The runs agree on that band and differ only in how wide the span is, so this is not counted as a disagreement.
- **split n/a vs verdict:** some runs classified the document as not a proposal and others graded it.
- **DISAGREE:** the labels differ and no single band is covered by every run, e.g. "Weak" in one run and "Strong" in another.
- **only 1 run / parse error:** too few readable reports to compare.

Per-paper verdict consistency: how many papers fall in each category.



Score spread per paper: each paper's highest run score minus its lowest, over papers with at least two scored runs.



Consistency aggregates

- all runs same verdict label: **309/585 (53%)** — of which 94 identical scores, 31 uniformly n/a
- different labels that all share a band (e.g. "Adequate" and "Adequate to Strong"): **202/585 (35%)**
- label disagreement, no band common to every run: **53/585 (9%)**
- split n/a vs a real verdict (triage disagreed between runs): **21**
- missing/unparseable runs: **0**

Category definitions are listed above the consistency chart.

Score spread within papers

Scores here are the rounded integers on the verdict line, so a spread of 1 can come from two runs whose exact scores were only slightly apart but rounded to neighboring integers. Max minus min of a paper's scores, over **547** papers with 2+ scored runs: mean **0.90**, median **1.0**, max **4**. Per-paper score stdev: mean **0.38**, max **1.41**.

Spread	Papers
0	142
1	323
2	76
3+	6

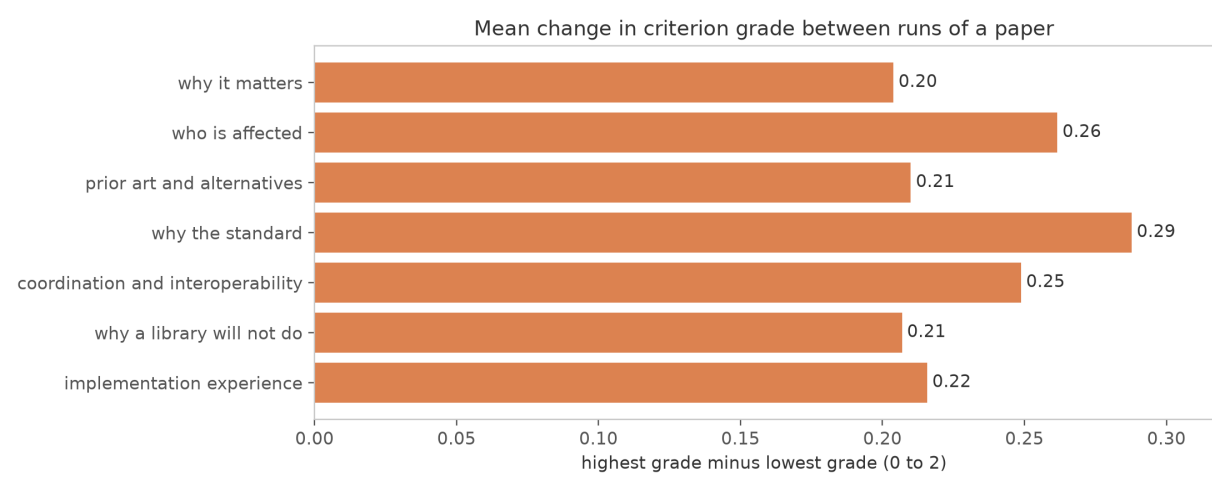
Sources of variation

The analyses below read the diagnostics block that the prompt appends to every report inside an HTML comment (`<!-- paperweight-diagnostics`
`... -->`), which is invisible when the report is rendered. It records the exact score before rounding, the grade of each criterion, every sample's vote on every section, and the total each sample would have produced on its own. Diagnostics were read from **2717** of the **2717** scored runs; n/a runs have none.

Which criteria move between runs

For each paper with at least two runs, each criterion's grade (0 to 2) is compared across the paper's runs. **Mean grade** is the average grade over all runs. **Mean change** is the average, over papers, of the criterion's highest grade minus its lowest. **Papers**

changed is the share of papers where the grade was not identical in every run, and **changed by 1+** is the share where it moved by a full point or more. The criteria with the largest changes contribute most to papers' scores and labels moving between runs.



Over **547** papers with at least two scored runs.

Criterion	Mean grade	Mean change	Papers changed	Changed by 1+
why it matters	1.59	0.20	50%	1%
who is affected	0.40	0.26	59%	5%
prior art and alternatives	1.59	0.21	51%	2%
why the standard	0.49	0.29	63%	4%
coordination and interoperability	0.42	0.25	54%	5%
why a library will not do	0.30	0.21	50%	3%
implementation experience	1.18	0.22	31%	9%

Whole-paper scores and band edges. The verdict line shows the score rounded to an integer, but the exact score is a fraction, and the rounded score crosses into the next band at exact scores of 0.5, 3.5, 7.5 and 11.5. A paper **changed band** if its runs' rounded scores fall in more than one band (span labels are ignored here), and a paper is **near an edge** if the mean of its exact scores is within 0.5 points of one of those values.

- mean change in exact score between runs: **0.92** points, against **0.89** for the rounded score
- papers that changed band: **132**; near an edge: **107/132 (81%)**
- papers in the same band every run: **415**; near an edge: **55/415 (13%)**

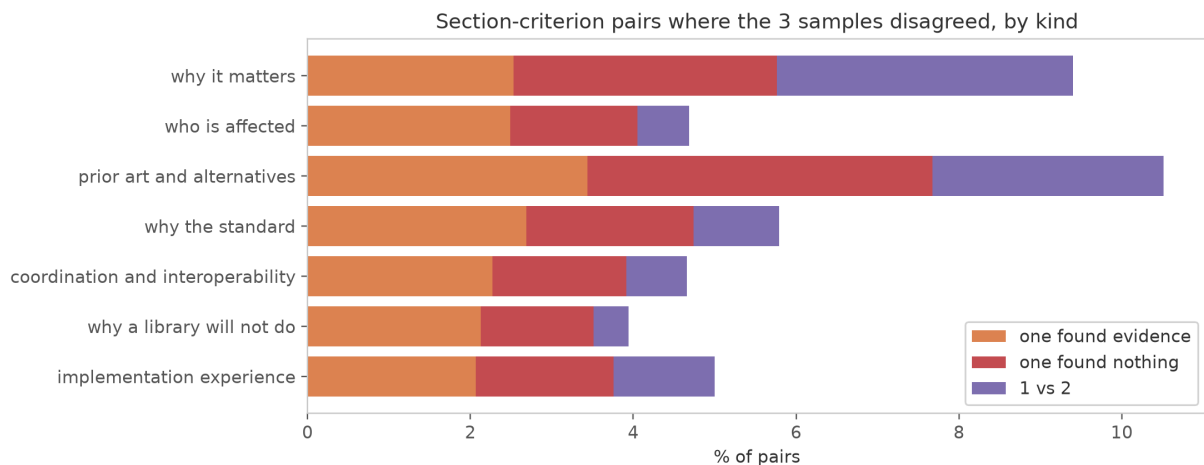
When most papers that changed band are near an edge, the band changes come mostly from small score differences landing on either side of an edge, not from large disagreements between runs.

Sample agreement and span labels

Every section is graded on every criterion by 3 independent samples. A section-criterion pair is **unanimous** when all 3 samples gave the same grade. A pair that is not unanimous is one of three kinds:

- **one found evidence:** one sample gave a non-zero grade and the other two gave 0, e.g. 0/0/1.
- **one found nothing:** one sample gave 0 and the other two found evidence, e.g. 2/2/0.
- **1 vs 2:** all three found evidence but disagreed on whether it was asserted (1) or supported (2), e.g. 1/2/2.

The first two are about whether a sample noticed a passage; only the third is about how strictly the scale was applied.



Criterion Pairs Unanimous One found evidence One found nothing 1 vs 2 why it matters
 28445 90.6% 719 921 1034 who is affected 28445 95.3% 707 445 181 prior art and
 alternatives 28445 89.5% 977 1206 807 why the standard 28445 94.2% 765 583 299
 coordination and interoperability 28445 95.3% 645 468 212 why a library will not do 28445
 96.1% 605 394 122 implementation experience 28445 95.0% 588 480 354

Where span labels come from. A span appears when the total a single sample would have produced on its own lands in a different band from the verdict. **595** of the **2717** scored runs (22%) received a span. For each of them, this counts how many of the 3 single-sample totals fell outside the verdict's band. A count of 1 means the span came from one sample disagreeing with the other two.

Samples outside the verdict's band	Span runs	Share
1	428	72%
2	145	24%
3	22	4%

Inconsistent papers — interactive dataframe

All 74 papers in the DISAGREE or split n/a vs verdict categories, sorted by score spread then score standard deviation, largest first. Papers with overlapping spans are not listed, because their runs share a band. **Spread** is the paper's highest run score minus its lowest, **stdev** is the population standard deviation of its run scores, and **verdicts** lists every run in run order as label(score). This is the sortable, paginated dataframe; a plain-text version of the same table follows below.

Q Search...					
☐	paper	category	spread	stdev	verdicts
☐	P4098R1	DISAGREE	3	1.26	Weak to Adequate(4) Weak(2) Weak(1) Weak to Adequate(4) Ade...
☐	P3242R2	DISAGREE	3	1.02	Adequate(5) Adequate(4) Adequate(5) Adequate(6) Weak(3)
☐	P3816R2	DISAGREE	3	1.02	Adequate(6) Adequate to Strong(7) Adequate to Strong(7) Stron...
☐	P4145R0	DISAGREE	2	0.98	Weak(3) Adequate(5) Adequate(5) Weak to Adequate(3) Adequat...
☐	P3899R1	DISAGREE	2	0.89	Strong(9) Strong(8) Adequate to Strong(7) Adequate(7) Strong(9)
☐	P2929R2	DISAGREE	2	0.75	Weak(2) Weak to Adequate(3) Adequate(4) Weak(3) Adequate(4)
☐	P3874R1	DISAGREE	2	0.75	Adequate to Strong(7) Adequate(7) Adequate to Strong(6) Stron...
☐	P3982R0	DISAGREE	2	0.75	Strong(8) Adequate(7) Adequate to Strong(7) Strong(9) Strong(8)
☐	P4139R0	DISAGREE	2	0.75	Weak(2) Weak to Adequate(3) Adequate(4) Weak(2) Weak(3)
☐	P4143R0	DISAGREE	2	0.75	Weak(2) Weak(3) Adequate(4) Weak(2) Weak to Adequate(3)

Inconsistent papers (74)

Same papers and columns as the dataframe above, as plain text.

Paper	Category	Spread	Stdev	Verdicts
P4098R1	DISAGREE	3	1.26	Weak to Adequate(4) Weak(2) Weak(1) Weak to Adequate(4) Adequate(4)
P3242R2	DISAGREE	3	1.02	Adequate(5) Adequate(4) Adequate(5) Adequate(6) Weak(3)
P3816R2	DISAGREE	3	1.02	Adequate(6) Adequate to Strong(7) Adequate to Strong(7) Strong(9) Strong(8)
P4145R0	DISAGREE	2	0.98	Weak(3) Adequate(5) Adequate(5) Weak to Adequate(3) Adequate(5)
P3899R1	DISAGREE	2	0.89	Strong(9) Strong(8) Adequate to Strong(7) Adequate(7) Strong(9)
P2929R2	DISAGREE	2	0.75	Weak(2) Weak to Adequate(3) Adequate(4) Weak(3) Adequate(4)
P3874R1	DISAGREE	2	0.75	Adequate to Strong(7) Adequate(7) Adequate to Strong(6) Strong(8) Adequate to Strong(8)
P3982R0	DISAGREE	2	0.75	Strong(8) Adequate(7) Adequate to Strong(7) Strong(9) Strong(8)
P4139R0	DISAGREE	2	0.75	Weak(2) Weak to Adequate(3) Adequate(4) Weak(2) Weak(3)
P4143R0	DISAGREE	2	0.75	Weak(2) Weak(3) Adequate(4) Weak(2) Weak to Adequate(3)
P4148R0	DISAGREE	2	0.75	Adequate(6) Strong(8) Adequate to Strong(7) Adequate(7) Adequate to Strong(8)
P4309R0	DISAGREE	2	0.75	Weak(3) Weak(3) Adequate(4) Adequate(4) Adequate(5)
P4318R1	DISAGREE	2	0.75	Strong to Excellent(11) Strong to Excellent(10) Strong to Excellent(11) Excellent(12) Strong(10)
P2929R3	DISAGREE	2	0.63	Weak to Adequate(3) Weak to Adequate(3) Weak(2) Weak(3) Adequate(4)
P3427R8	DISAGREE	2	0.63	Adequate(6) Adequate(7) Strong(8) Adequate to Strong(7) Adequate to Strong(7)
P3818R2	DISAGREE	2	0.63	Adequate(6) Adequate to Strong(7) Adequate to Strong(7) Strong(8) Adequate(7)

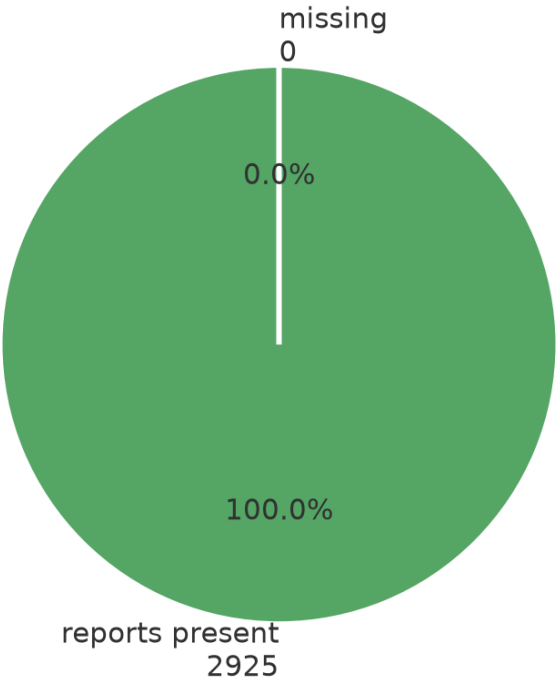
Paper	Category	Spread	Stdev	Verdicts
P3864R1	DISAGREE	2	0.63	Weak(3) Adequate(4) Weak(2) Weak(3) Weak(3)
P3876R1	DISAGREE	2	0.63	Adequate to Strong(7) Adequate to Strong(7) Adequate(6) Strong(8) Adequate to Strong(7)
P4124R0	DISAGREE	2	0.63	Adequate to Strong(7) Adequate to Strong(7) Adequate to Strong(7) Strong(8) Adequate(6)
P4318R0	DISAGREE	2	0.63	Strong to Excellent(11) Excellent(12) Strong to Excellent(11) Strong(10) Strong to Excellent(11)
P4035R1	split n/a vs verdict	1	0.50	Adequate(6) Adequate to Strong(7) n/a Adequate(7) Adequate(6)
P4208R0	split n/a vs verdict	1	0.50	n/a Weak(2) n/a Weak to Adequate(3) n/a
P1040R11	DISAGREE	1	0.49	Strong to Excellent(11) Excellent(12) Strong(11) Strong to Excellent(12) Excellent(12)
P2956R3	DISAGREE	1	0.49	Weak to Adequate(3) Weak(3) Weak to Adequate(3) Weak to Adequate(4) Adequate(4)
P3642R5	DISAGREE	1	0.49	Adequate to Strong(7) Adequate(7) Strong(8) Strong(8) Adequate to Strong(7)
P3724R4	DISAGREE	1	0.49	Adequate to Strong(7) Adequate to Strong(7) Strong(8) Adequate to Strong(8) Adequate(7)
P3904R2	DISAGREE	1	0.49	Adequate(7) Adequate to Strong(8) Strong(8) Adequate to Strong(7) Adequate to Strong(7)
P3928R1	DISAGREE	1	0.49	Weak(3) Weak to Adequate(3) Adequate(4) Adequate(4) Adequate(4)
P3982R2	DISAGREE	1	0.49	Adequate(7) Adequate to Strong(7) Adequate to Strong(7) Strong(8) Adequate to Strong(8)
P4030R1	DISAGREE	1	0.49	Weak to Adequate(3) Weak(3) Adequate(4) Weak to Adequate(4) Weak to Adequate(3)
P4090R1	DISAGREE	1	0.49	Adequate(7) Adequate to Strong(7) Adequate to Strong(8) Strong(8) Adequate to Strong(7)
P4136R0	DISAGREE	1	0.49	Adequate to Strong(7) Strong(8) Strong(8) Adequate(7) Adequate to Strong(8)
P4136R2	DISAGREE	1	0.49	Adequate(7) Adequate to Strong(7) Adequate to Strong(7) Strong(8) Adequate to Strong(8)
P4147R0	DISAGREE	1	0.49	Weak to Adequate(3) Adequate(4) Weak to Adequate(3) Adequate(4) Weak(3)
P4157R0	DISAGREE	1	0.49	Weak(3) Weak to Adequate(3) Weak to Adequate(4) Adequate(4) Adequate(4)

Paper	Category	Spread	Stdev	Verdicts
P4304R0	DISAGREE	1	0.49	Adequate(4) Weak(3) Weak to Adequate(3) Weak to Adequate(4) Weak(3)
P4025R2	split n/a vs verdict	1	0.43	Adequate(5) Adequate(5) n/a Adequate(4) Adequate(5)
P4097R1	split n/a vs verdict	1	0.43	Weak to Adequate(2) n/a Weak to Adequate(3) Weak(2) Weak(2)
P4137R0	split n/a vs verdict	1	0.43	n/a Adequate(5) Adequate(5) Adequate(4) Adequate(5)
P4178R0	split n/a vs verdict	1	0.43	n/a Weak to Adequate(3) Weak to Adequate(4) Weak to Adequate(3) Weak to Adequate(3)
P4317R1	split n/a vs verdict	1	0.43	Strong(10) Strong(10) n/a Strong(9) Strong(10)
P1130R2	DISAGREE	1	0.40	Adequate(4) Weak(3) Adequate(4) Adequate(4) Adequate(4)
P3097R2	DISAGREE	1	0.40	Strong(8) Adequate to Strong(8) Adequate to Strong(8) Adequate(7) Strong(8)
P3400R3	DISAGREE	1	0.40	Adequate(7) Adequate(7) Strong(8) Adequate to Strong(7) Adequate to Strong(7)
P3440R3	DISAGREE	1	0.40	Strong(8) Strong(8) Strong(8) Adequate(7) Strong(8)
P3950R0	DISAGREE	1	0.40	Adequate(7) Adequate(7) Adequate(7) Strong(8) Adequate to Strong(7)
P3952R0	DISAGREE	1	0.40	Adequate(7) Strong(8) Adequate to Strong(7) Adequate to Strong(7) Adequate to Strong(7)
P3973R1	DISAGREE	1	0.40	Adequate(7) Adequate to Strong(7) Strong(8) Adequate to Strong(7) Adequate to Strong(7)
P4008R0	DISAGREE	1	0.40	Weak(3) Weak(3) Adequate(4) Weak(3) Weak(3)
P4015R0	DISAGREE	1	0.40	Weak to Adequate(3) Weak to Adequate(3) Weak(3) Adequate(4) Weak to Adequate(3)
P4030R0	DISAGREE	1	0.40	Weak to Adequate(3) Adequate(4) Weak(3) Weak(3) Weak to Adequate(3)
P4140R0	DISAGREE	1	0.40	Adequate(4) Weak(3) Weak to Adequate(3) Weak to Adequate(3) Weak(3)
P4148R2	DISAGREE	1	0.40	Adequate to Strong(7) Adequate to Strong(7) Strong(8) Adequate(7) Adequate(7)
P4156R0	DISAGREE	1	0.40	None to Weak(0) None to Weak(0) None to Weak(0) None(0) Weak(1)

Paper	Category	Spread	Stdev	Verdicts
P4213R0	DISAGREE	1	0.40	Adequate(7) Adequate to Strong(7) Strong(8) Adequate to Strong(7) Adequate to Strong(7)
P4222R2	DISAGREE	1	0.40	Adequate to Strong(7) Adequate(7) Adequate to Strong(7) Adequate to Strong(7) Strong(8)
P4227R0	DISAGREE	1	0.40	Weak to Adequate(3) Weak(3) Adequate(4) Weak to Adequate(3) Weak to Adequate(3)
P4231R0	DISAGREE	1	0.40	Weak to Adequate(3) Weak to Adequate(3) Weak(3) Adequate(4) Weak to Adequate(3)
P4282R1	DISAGREE	1	0.40	Weak(3) Weak to Adequate(3) Adequate(4) Weak to Adequate(3) Weak(3)
P4311R1	DISAGREE	1	0.40	Strong to Excellent(11) Strong(11) Strong(11) Strong to Excellent(11) Excellent(12)
P1000R8	split n/a vs verdict	0	0.00	n/a n/a Weak(1) n/a n/a
P2000R5	split n/a vs verdict	0	0.00	n/a n/a n/a Adequate(4) n/a
P3873R0	split n/a vs verdict	0	0.00	n/a None(0) n/a n/a n/a
P3965R0	split n/a vs verdict	0	0.00	None(0) None(0) n/a None(0) None(0)
P3970R0	split n/a vs verdict	0	0.00	Adequate to Strong(8) n/a n/a Adequate to Strong(8) n/a
P4007R2	split n/a vs verdict	0	0.00	Weak to Adequate(3) Weak(3) Weak(3) n/a n/a
P4024R0	split n/a vs verdict	0	0.00	Weak(1) n/a n/a n/a n/a
P4025R1	split n/a vs verdict	0	0.00	n/a Adequate(5) Adequate(5) n/a Adequate(5)
P4027R0	split n/a vs verdict	0	0.00	None(0) None(0) n/a n/a None(0)
P4125R1	split n/a vs verdict	0	0.00	Adequate(7) n/a n/a n/a n/a
P4225R0	split n/a vs verdict	0	0.00	n/a None(0) n/a n/a None(0)
P4330R0	split n/a vs verdict	0	0.00	n/a n/a n/a n/a Strong(8)

Paper	Category	Spread	Stdev	Verdicts				
P5000R0	split n/a vs verdict	0	0.00	n/a	Weak(2)	n/a	n/a	n/a
P5000R1	split n/a vs verdict	0	0.00	n/a	n/a	Weak(2)	Weak(2)	n/a

Expected runs (2925 = 585 papers x 5 runs)



Missing runs

All expected reports are present. Expected paper list from the paperstore DB (585 papers); 5 runs per paper (from max run number).